James Schaffer jamesschaffer.dev@gmail.com Linkedin : james-schaffer-dev

Personal Profile

Second year student at University of Leicester studying Computer Science (1st class, 80%). Passionate about software engineering, game development, automation and data handling. Have worked in a team and independently on many projects and competitions. Lead teams to winning and getting 2nd place in multiple hackathons, some sponsored by the likes of IBM and Capital One. Taken the initiative to teach myself multiple game engines and 4 programming languages to develop games, on of which was nominated for BAFTA YGD's Game making award in 2023.

BAFTA YDG Finalist, 2x Second place hackathons, IBM rookie innovator award, Morgan Stanley Coding Challenge Second place

Projects and other experience

Employment and volunteering

Bartender, Bank of Beers July 2025 - Present

- Worked in a fast paced, customer-facing environment, where I'm always learning on the job.
- Have the responsibilities of handling and balancing keeping fridges stocked, glasses clean and serving customers.

Hackathons / coding challenges :

Capital One x IBM LetsHack Leicester Hackathon 2025 (**2nd place**) - FindMyParking (Team of 5)

- Enable commuting students to find cheap parking nearby and, by linking their outlook calendar, plan their day
- Worked on designing the concept and app as well as coding interactive map features, frontend and backend systems, automated test data generation. Self taught Next.js and Typescript during development
- Used : Next js, Typescript, Leaflet js, Open street maps API, Google API, Outlook API, Tailwindcss, HTML/CSS, Git&Github

LeicesterCS Coding Challenge 2025 (**2nd place**) (Team of 4)

- Worked on 2 problems, one involving finding voxel pyramid heights based on a number of voxels and another which used hashing to find duplicate files in a file system. Leetcode and Advent of Code experience helped, resulting in the team achieving 2nd place.
- Used : Python

IBM Hackathon 2025 (**Won rookie innovator award**) - MyBudget (Team of 4)

- We created a budgeting web app which categorises purchases. Users can take pictures of their receipts, the app then scans the contents and then sorts the purchases into categories. This system allows for users to manually input purchases, meaning it could track any purchase made by them, not just those made with a specific credit card, like in other similar solutions.
- I did the frontend design and linking the frontend and backend.
- Used : Javascript, HTML/CSS, Bootstrap, Flask, Python, SQLite3, Gemini AI API

Capital One LetsHack Leicester Hackathon 2024 - Choremate (Team of 5)

- We developed a chore management mobile app tailored to university students. It dynamically allocates them in user-defined groups based on a rota and adapts to people missing chores, switching or doing extra to make it as fair as possible.
- I worked on the back end, building the database and a python backend.
- Using : Python, Kivy, Firebase

Morgan Stanley coding challenge 2024 (**2nd place**)

- In a team we designed and developed an algorithm to automatically trade in a simulated FX market over a period of 2 hours, I designed and implemented an algorithm using 2 trends to take into account short and long term trends to best predict the market.
- Using : python, morgan stanley's API

Game Jams 2023 & 2024

- I have completed 2 game jams (Brackeys Game Jam 2024.2 & 2023.1). I made all the code, art and sound assets as well as designing the game, and received positive feedback from players.
- Using : (Unity, c#, Aseprite) & (Godot, gdScript, Aseprite)

Projects :

Colour Space conversion and image management system 2025

- Creating a colour space and image management system: handles converting between multiple colour spaces such as sRGB, OKLab, XYZ and HSV; image import/export/generation, allows for UV mappings.
- Originally built in python then c# and finally moved to c++ to optimise for larger images and converting between colour spaces, the original system used for perlin noise and colour space comparisons.
- Using : C++, Cimg
- Github link : github.com/James-Schaffer/imageManager

Portfolio Website 2023-2025

- Created a personal portfolio website to showcase projects, first in 2023 after BAFTA YGD nomination, using HTML, CSS & JS and hosted using google cloud with cloudflare

- Re-developed in 2025 and hosted on github pages with cloudflare
- Using : HTML, CSS, JS, Github pages, Google cloud (old site), Cloudflare
- Website link : www.jamesschaffer.dev
- Github link : github.com/James-Schaffer/James-Schaffer.github.io

Random distribution & Cantor paring research project 2025
- Project researching, developing and benchmarking different random point distribution algorithms and mapping infinite 2D arrays to 1D arrays (cantor paring). Built benchmarking tools in python to test the different algorithms
- Using : Python, Matplotlib
- Link : jamesschaffer.dev/html/portfolio/programming/mappingInfinite2DArraysToA1DArray.html

Recipe timer application 2024 - A-level coursework
- Developed a mobile application and wrote a full report of the project involving the analysis, research, design and development processes
- Provided the user with a recipe sharing platform with built in cooking support in the form of an responsive timer adapting to how long you take on each part of the recipe
- Implemented an account system which allowed users to personalise their experience and find new recipes
- Using : Python, Kivy, SQLite
- Link to project and reflection : jamesschaffer.dev/html/portfolio/programming/recipeManagerApp.html

Escape the darkness 2023 (**BAFTA YGD FINALLIST**)
- Developed as short 2D puzzle-survival game, nominated for **BAFTA YGD 2023** 15-18 game making category, featured on local news
- Most code and all art assets created by me. Built many core systems including a real-time FOV mask which enabled many of the main gameplay mechanics to function
- Using : Unity, C#, Aseprite
- Link : jamesschaffer.dev/html/portfolio/games/escapeTheDarkness.html

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Technical skills
Languages : Python, Java, C#, C++, Javascript, GDScript

Frameworks/Tools : Next.js, React, Godot, Unity, Flask, Leaflet

APIs : Google API, OpenStreetMap, Outlook API

Soft skills
Teamwork, communication, time management, problem solving, attention to detail, reliability

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Education
University of Leicester : September 2024-Present

BSc (Hons) Computer Science with year in Industry

Completed 1st year with an overall mark of 80% (**equivalent to a 1st**)
- Year 1 modules : Computer Fundamentals , Programming Fundamentals , Mathematical Fundamentals , Computer Architecture , Introduction to Object Oriented Programming , Requirements Engineering and Professional practice , Algorithms, Data Structures and Advanced Programming , Foundations of Computation

Acting course rep for second year computer science : September 2025 - present
- Acting as a voice and representative for year 2 computer science students
- Gathering feedback from students and passing it to appropriate people/groups to incite positive change and pass back information to the students

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Achievements & Licences
- **Capital One x IBM** LetsHack Leicester Hackathon 2025 **2nd place**
- **IBM** MQ Developer essentials - 2025
- **Rookie innovator award** and **finalist** at **IBM** x Leicester CS society hackathon 2025
- **Morgan Stanley** Leicester Coding Challenge 2024 **2nd place**
- **Gold DofE** - 7/9/2019 - 25/03/2024
- 2023 **BAFTA YGD Finalist** - Escape The Darkness

Interests
- Computer game development (4 complete games) & Computer graphics : gradient noise, colour theory
- Computer programming (many projects in areas such as : procedural generation, life simulations, web development)
- Member of Computer Science Society in Leicester
- Volunteering : Whilstlewood common - community garden/permaculture center, Radio museum of Great Britain, libraries